

- 1 The olive plant, *Olea europaea*, is grown in many parts of the world. The fruits of the plant (olives) and the oil that can be obtained from the fruits (olive oil), provide food for humans.

Triglycerides are the main type of lipid in olive oil. They are synthesised in the olive plant from glycerol and fatty acids.

Scientists can analyse samples of different olive oils to identify:

- the fatty acids used to synthesise triglycerides
- the composition of the different triglycerides present.

- (a) Table 1.1 shows some details of the five most common fatty acids found in samples of olive oil produced by olive plants grown in different regions in Portugal.

Table 1.1

Key

C : D = number of carbon atoms : number of double bonds in the hydrocarbon chain

X = missing detail

fatty acid	percentage of total fatty acid content	C : D	chemical structure
oleic acid	55.0–83.0	18 : 1	$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{X}$
palmitic acid	7.5–20.0	16 :	$\text{CH}_3(\text{CH}_2)_{14}\text{X}$
linoleic acid	3.5–21.0	18 :	$\text{CH}_3(\text{CH}_2)_4(\text{CH}=\text{CH})\text{CH}_2(\text{CH}=\text{CH})(\text{CH}_2)_7\text{X}$
stearic acid	0.5–5.0	18 :	$\text{CH}_3(\text{CH}_2)_{16}\text{X}$
palmitoleic acid	0.3–3.5	16 :	$\text{CH}_3(\text{CH}_2)_5\text{CH}=\text{CH}(\text{CH}_2)_7\text{X}$

- (i) Table 1.1 shows the **C : D** values for oleic acid.

In Table 1.1, write the values for **D** for each of the four other fatty acids listed. [1]

- (ii) In the first column of Table 1.1, draw a circle around each of the fatty acids that can be described as saturated. [1]

- (iii) State the detail of chemical structure, represented by **X**, which is missing from Table 1.1.

..... [1]

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(iv) The analysis of the triglycerides present in the different samples of olive oil showed that:

- there are many different triglycerides present in olive oil
- each olive oil is different in its composition, but the same few triglycerides are present in all olive oils.

With reference to Table 1.1 and to the structure of triglycerides, suggest explanations for these observations.

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..... [2]

(b) Glycerol is soluble in water. Triglycerides are insoluble in water.

Explain why water is a good solvent for some substances such as glycerol, but is a poor solvent for substances such as triglycerides.

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..... [2]

(c) Phloem is the plant tissue responsible for the transport of organic substances, such as fatty acids, from one area of a plant to another. The tissue is composed of more than one type of cell.

Name the type of cell that forms the transport vessels of phloem tissue.

..... [1]

[Total: 8]